

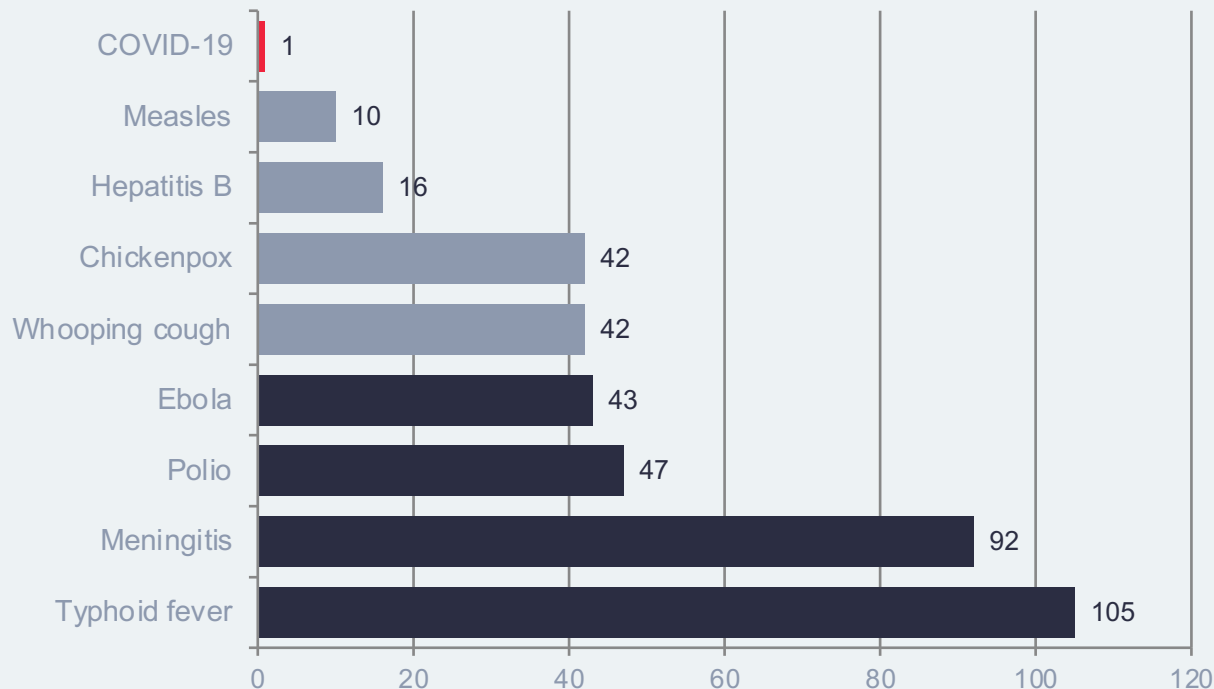
The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

1796 – 2020 | Two Centuries of Immunization Progress

Source: Our World in Data · ourworldindata.org/vaccination

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating



< 1 YEAR

from pathogen
identification to licensure

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

US Vaccine-Preventable Disease Burden: Pre-Vaccine vs Post-Vaccine Era

Disease	Pre-Vaccine Cases/million/yr	Post-Vaccine Cases/million/yr	Case Reduction	Pre-Vaccine Deaths/million/yr	Death Reduction
Diphtheria ✓	158	0	100%	13.7	100%
Smallpox ✓	250	0	100%	2.9	100%
Acute polio ✓	141	0	100%	10	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Rubella	242	0.04	99.98%	0.09	100%
Measles	3,044	0.2	99.99%	2.5	100%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

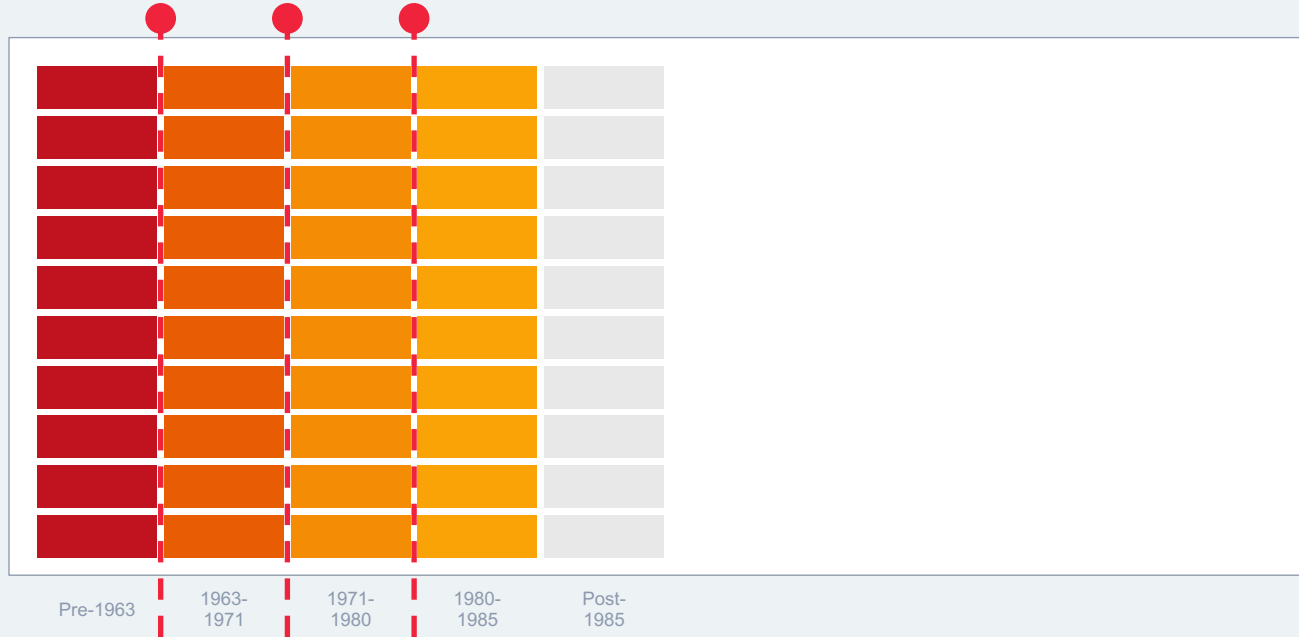
● 100% Case Reduction Achieved

** Pneumococcal disease still presents a remaining challenge with only 40.5% case reduction*

Source: Roush & Murphy (2007) · CDC data via Our World in Data

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Incidence per 100,000 population by state group, 1950-1990



196

3

First measles vaccine developed by John Enders

197

1

MMR vaccine introduced by Maurice Hilleman

198

0

Kindergarten vaccination mandate enacted

Key Statistics

100-1,000+

cases/100K pre-1963



~0

cases/100K post-1980

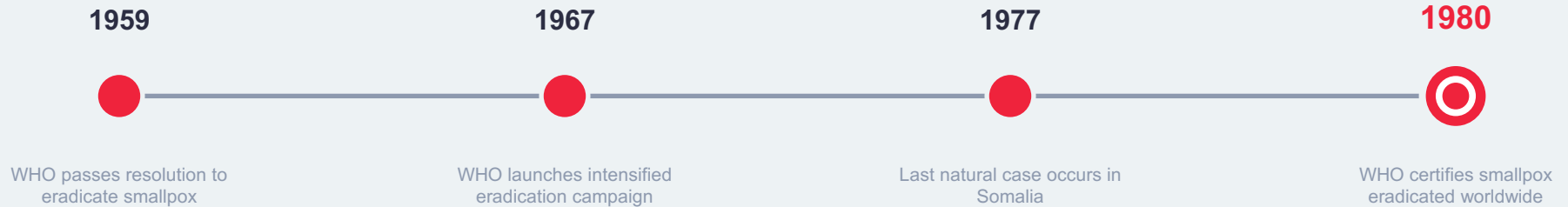


High Incidence

Near Zero

Source: CDC measles surveillance data via Our World in Data

Smallpox: The Only Human Disease Eradicated Through Vaccination



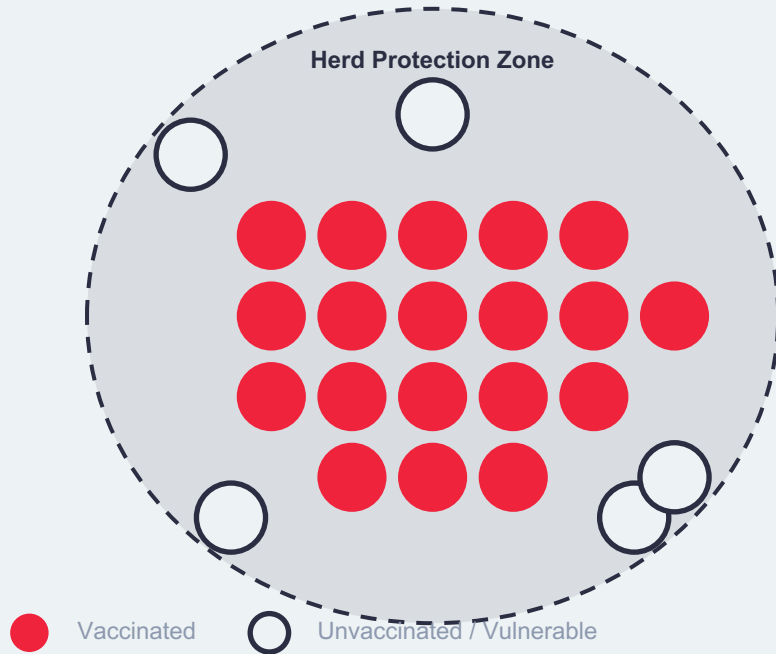
Why smallpox?

The virus infected only humans, symptoms were distinguishable, and the vaccine was cheap and effective with long-lasting immunity.

Source: WHO · Our World in Data

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Vaccination creates collective protection that shields the most vulnerable



Protected Groups

- Newborns
- Immunocompromised individuals
- Cancer patients on chemotherapy
- Infants too young to vaccinate

Key Insight

When enough people are vaccinated, pathogens cannot spread effectively — protecting those who cannot be vaccinated.

~80%

vaccination coverage

5-10%

cannot be vaccinated

Gaps Persist: Some Diseases Show Incomplete Vaccine Impact

40.5%

case reduction

Pneumococcal Disease

31.3% death reduction

89%

case reduction

Hepatitis A

83.9%

case reduction

Acute Hepatitis B



Access Gaps

Not everyone globally has equal access to vaccines



Supply Shortages

Vaccine production cannot always meet demand



Trust Challenges

Poor scientific understanding can reduce uptake

Source: Roush & Murphy (2007) · Our World in Data

Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 1 100% Elimination Achieved** — Vaccines eliminated 100% of cases and deaths for diphtheria, smallpox, and polio in the US
- 2 Development timelines collapsed** — From 105 years (typhoid) to under 1 year (COVID-19), driven by mRNA and genomic tools
- 3 Measles eliminated nationwide** — Cases dropped to near zero across all 50 US states following the 1980 kindergarten mandate
- 4 Smallpox — the only eradication** — Certified in 1980 after a global campaign; remains the only human disease fully eradicated
- 5 Gaps persist** — Pneumococcal (40.5%), hepatitis A (89%), and hepatitis B (83.9%) show incomplete control

Improve access, coverage, and trust in vaccination — because the job isn't done.